Orifice Flow

Clint Guymon, PhD PE

Brigham Young University

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Learning Outcomes

- Estimate flow through an orifice for incompressible liquids, compressible gases (choked and unchoked flow), and two-phase flow for flashing fluids.
- Understand the application of the mechanical energy balance to calculate fluid velocity and pressure drops.
- Identify appropriate discharge coefficients for different fluid types to account for frictional losses

Reading

- Foundations of Spiritual and Physical Safety: with Chemical Processes; Chapter VI.3.2 (Orifice Flow) through Sec. VI.4.

1 Mechanical Energy Balance

Note
Constant Density, Frictionless, Constant Temperature Flow

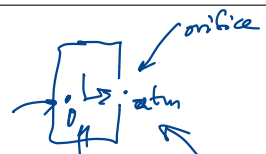
friction

$$\frac{\Delta P}{\rho g} + \frac{\Delta u^2}{2g} + \Delta z + \frac{F}{g} = \frac{-W_s}{\dot{m} g} \tag{1}$$

where ΔP is the change in pressure, Δu is the change in velocity, Δz is the change in elevation, ρ is density, F is the frictional losses, and W_s is the shaft work with respect to two points in the system.

If there is no friction and constant density and no shaft work and no change in elevation, the mechanical energy balance simplifies to:

Note



$$\frac{\Delta P}{\rho g} + \frac{\Delta u^2}{2g} = 0 \tag{2}$$

or assuming that the flow points are inside a tank where P = P_o and u = 0 and outside the tank where P = P_atm and u = u, then

$$\frac{P_o}{\rho} - \frac{P_{atm}}{\rho} = \frac{u^2}{2} \tag{3}$$

or

$$u = \sqrt{2\left(\frac{P_o - P_{atm}}{\rho}\right)} \tag{4}$$

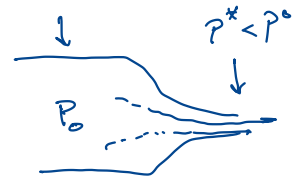
$\mu A = F, Q$

incompressible

2 Incompressible Flow through an Orifice

Thus for incompressible flow, the velocity of the fluid can be determined from the pressure drop across an orifice. Although the flow is usually not frictionless, the velocity can be estimated from the pressure drop and the density of the fluid according to:

$$\dot{m} = AC_d\sqrt{2\rho\Delta P} \tag{5}$$



or

$$\dot{n} = \frac{AC_d}{M_w}\sqrt{2\rho\Delta P} \tag{6}$$

where $\dot{m}$ is the mass flow rate, $\dot{n}$ is the molar flow rate, M_w is the molecular weight, A is the area of the orifice, ρ is the density of the fluid, C_d is the discharge coefficient, and ΔP is the pressure drop across the orifice. For liquid flow, and estimate of the discharge coefficient is 0.65.

3 Gaseous flow through an Orifice: General Case

The flow of a gas through an orifice is also derived from an energy balance assuming adiabatic and frictionless flow. The flow is characterized by the pressure ratio, P_o/P_{atm} and associated Mach number, Ma. P_o is the absolute pressure of the volume. The Mach number can be found from:

$$Ma = \min\left(1, \sqrt{\frac{2}{\gamma - 1}\left(\left(\frac{P_o}{P_{atm}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right)}\right) \tag{7}$$

~2, 1.8

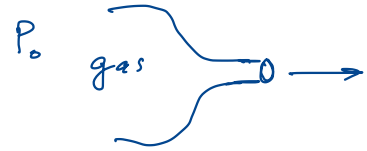
where γ is the ratio of specific heats (1.4 for air). The flow is choked when $Ma = 1$ and unchoked when $Ma < 1$.

The molar flow rate is given by:

$$\dot{n} = P_o AC_d\sqrt{\frac{\gamma}{RTM_w}}Ma\left[1 + \frac{(\gamma - 1)}{2}Ma^2\right]^{\frac{\gamma+1}{2-2\gamma}} \tag{8}$$

where R is the gas constant and T is the temperature, P_o is the absolute pressure inside the vessel, C_d is the discharge coefficient (typically about 0.9). If you use all SI units (Pascal, meters, kilograms, mole: so pressure in pascal, area in meters, R in J/mol/K, and Mw in kg/mol), you'll get mole per second for $\dot{n}$. See Perry's Handbook for Chemical Engineers Equation 6-118 for more information.

4 Flashing Two Phase Flow Estimate

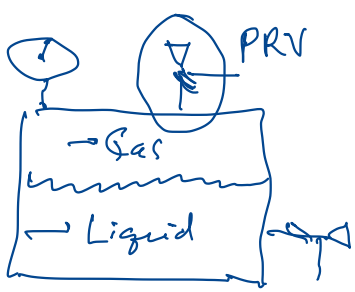


$$\dot{m} = f\frac{\Delta H_v A}{\nu_{fg}}\sqrt{\frac{1}{C_p T_s}} \tag{9}$$

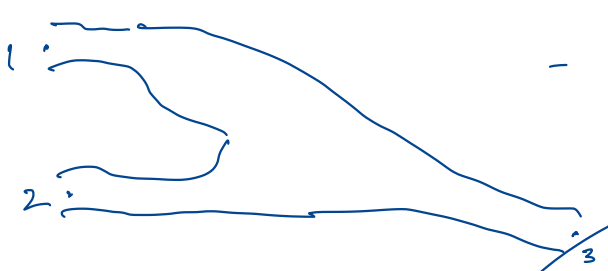
where $\dot{m}$ is the mass flow rate, f is a factor depending on the scenario (less than 1), ΔH_v is the heat of vaporization, A is the area of the orifice, ν is the specific volume change of the liquid as it flashes (ν_g - ν_l), C_p is the heat capacity of the fluid, and T_s is the saturation temperature of the fluid at the set pressure. See Chapter 4 of Crowl and Louvar for more information.

Action Items

1. Compare Eq. VI.27(Guymon, 2025) to the mechanical energy balance equation in your fluid mechanics textbook. Explain how they agree or disagree including differences in perhaps the molar volume versus density and the molar flow rate and the mass flow rate. What assumptions are made in the derivation?
2. Derive the mechanical energy balance if we had two inlet fluid flows (instead of one as is typically used in the mechanical energy balance) into the control volume and one fluid flow out.



$$\frac{dU}{dt} = 0 = \dot{n}^1_{in}\left[P_1\bar{V} + \frac{v_1^2}{2}M_w + gz_1M_w\right] + \dot{n}^2_{in}\left[P_2\bar{V} + \frac{v_2^2}{2}M_w + gz_2M_w\right] - \dot{n}^3_{out}\left[P_3\bar{V} + \frac{v_3^2}{2}M_w + gz_3M_w\right] + \cancel{\dot{Q}}^{\,0} + \dot{W}_s$$



$$\bar{H} = \bar{U} + P\bar{V}$$

$$\dot{n}^1_{in}\bar{U}_1 + \dot{n}^2_{in}\bar{V}_2 = \dot{n}^3_{out}\bar{V}_3$$

$$\dot{n}^1_{in} + \dot{n}^2_{in} = \dot{n}^3_{out}$$

References

C. Guymon. *Foundations of Spiritual and Physical Safety: with Chemical Processes*. 2025.

$$\frac{dn\bar{U}}{dt} = \sum_{in} \dot{n}_{in}\left[\bar{H} + \frac{V^2 M_w}{2} + gzM_w\right] - \sum_{out} \dot{n}_{out}\left[\bar{H} + \frac{V^2 M_w}{2} + gzM_w\right] + \dot{Q}_{CV} + \dot{W}_{s\,CV}$$

Steady State $\frac{dnU}{dt} = 0$

Incompressible $\rho = \text{const}$, $\bar{V} = \text{const.}$ $\leftarrow \dot{n}_{in} = \dot{n}_{out}$

No reactions

Const. Temperature

$$\bar{H} = \bar{U} + P\bar{V}$$

$$0 = \dot{n}\left[\cancel{\bar{U}_1} + P_1\bar{V} + \frac{V_1^2}{2}\cdot M_w + gz_1 M_w\right] - \dot{n}\left[\cancel{\bar{U}_2} + P_2\bar{V} + \frac{V_2^2}{2} M_w + gz_2 M_w\right] + \cancel{\dot{Q}}^{\,0} + \dot{W}_s$$

$\bar{V}[=]\ \frac{m^3}{mol}$ $\quad \bar{V}/M_w = \frac{1}{\rho}$

$$0 = \cancel{\dot{m}}\left[\cancel{u_1} + P_1\bar{v} + \frac{V_1^2}{2}\cdot M_w + g z_1 M_w\right]$$

$$- \cancel{\dot{m}}\left[\cancel{u_2} + P_2\bar{v} + \frac{V_2^2}{2} M_w + g z_2 M_w\right] + \cancel{\dot{Q}}^{\,0} + \frac{\dot{W}_s}{\dot{m}}$$

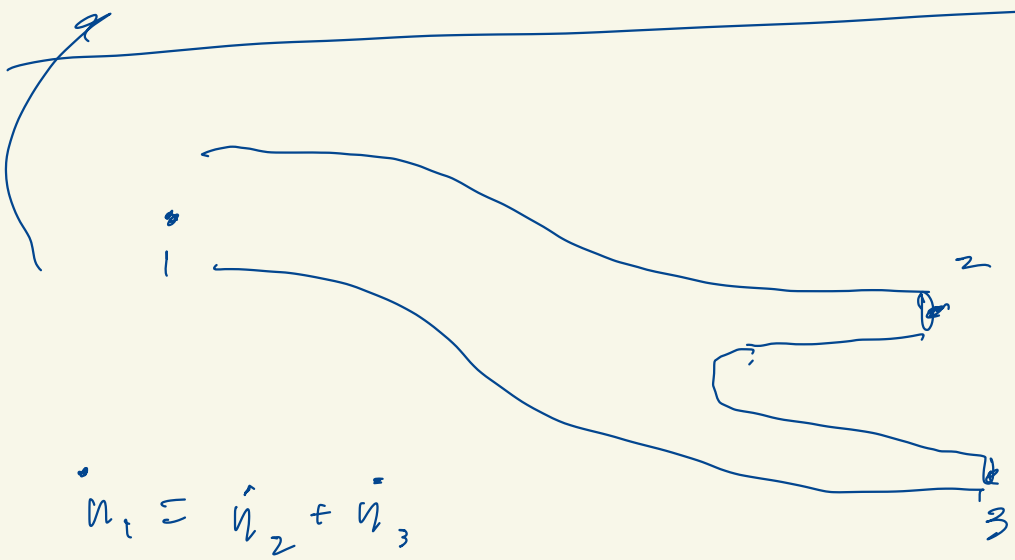
No heat input or flow out

→ adiabatic

$$\left(\frac{P_1}{\rho}\cdot \cancel{M_w} + \frac{V_1^2}{2}\cancel{M_w} + g z_1 \cancel{M_w} - \frac{P_2}{\rho}\cancel{M_w} + \frac{V_2^2}{2}\cancel{M_w} + g z_2 \cancel{M_w}\right) + \frac{\dot{W}_s}{\dot{m} M_w} = 0$$

$\dot{m}$

$$\frac{\Delta P}{\rho} + \frac{\Delta V^2}{2} + g\Delta z = -\frac{\dot{W}_s}{\dot{m}} \qquad \Delta P = P_1 - P_2$$



$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3$$